

VELORA — Module Inventory (Part 1)

Migrations · Enums · Models

Folder Structure

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```

1. Migrations

Warehouses Table

```
<?php
// database/migrations/inventory/2026_05_19_000030_create_warehouses_table.php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    public function up(): void
    {
        Schema::create('warehouses', function (Blueprint $table) {
            $table->uuid('id')->primary();
            $table->uuid('business_id');
            $table->uuid('branch_id')->nullable()->comment('Terkait dengan cabang mana');

            $table->string('name', 100);
            $table->string('code', 20)->nullable();
            $table->string('address')->nullable();
            $table->boolean('is_main')->default(false);
            $table->boolean('is_active')->default(true);

            $table->timestamps();
            $table->softDeletes();

            $table->foreign('business_id')->references('id')->on('businesses')->cascadeOnDelete();
            $table->foreign('branch_id')->references('id')->on('branches')->nullOnDelete();

            $table->unique(['business_id', 'code']);
            $table->index(['business_id', 'is_active', 'is_main']);
        });
    }

    public function down(): void { Schema::dropIfExists('warehouses'); }
};
```

Stock Movements Table

```
<?php
```

```
// database/migrations/inventory/2026_05_19_000031_create_stock_movements_table.php
```

```
use Illuminate\Database\Migrations\Migration;
```

```
use Illuminate\Database\Schema\Blueprint;
```

```
use Illuminate\Support\Facades\Schema;
```

```
return new class extends Migration
```

```
{
```

```
    public function up(): void
```

```
    {
```

```
        Schema::create('stock_movements', function (Blueprint $table) {
```

```
            $table->uuid('id')->primary();
```

```
            $table->uuid('business_id');
```

```
            $table->uuid('warehouse_id');
```

```
            $table->uuid('user_id')->comment('Siapa yang membuat movement');
```

```
            $table->string('reference_number', 50)->unique();
```

```
            $table->string('type', 20)->comment('in, out, transfer, adjustment, opname, return');
```

```
            $table->string('status', 20)->default('completed')->comment('draft, pending, completed');
```

```
            $table->text('notes')->nullable();
```

```
            // Referensi ke transaksi lain (misal ID Sales / Purchase Order)
```

```
            $table->uuid('reference_id')->nullable();
```

```
            $table->string('reference_type')->nullable(); // misal 'sales', 'purchases'
```

```
            $table->timestamps();
```

```
            $table->foreign('business_id')->references('id')->on('businesses')->cascadeOnDelete();
```

```
            $table->foreign('warehouse_id')->references('id')->on('warehouses');
```

```
            $table->foreign('user_id')->references('id')->on('users');
```

```
            $table->index(['business_id', 'type', 'status']);
```

```
            $table->index(['business_id', 'created_at']);
```

```
        });
```

```
    }
```

```
    public function down(): void { Schema::dropIfExists('stock_movements'); }
```

```
};
```

Stock Movement Items Table

```
<?php
```

```
// database/migrations/inventory/2026_05_19_000032_create_stock_movement_items_table.php
```

```
use Illuminate\Database\Migrations\Migration;
```

```
use Illuminate\Database\Schema\Blueprint;
```

```
use Illuminate\Support\Facades\Schema;
```

```
return new class extends Migration
```

```
{
```

```
    public function up(): void
```

```
    {
```

```
        Schema::create('stock_movement_items', function (Blueprint $table) {
```

```
            $table->uuid('id')->primary();
```

```
            $table->uuid('stock_movement_id');
```

```
            $table->uuid('product_id');
```

```
            $table->uuid('product_variant_id')->nullable();
```

```
            $table->integer('quantity'); // Positif/Negatif akan disesuaikan saat hitung, di sir
```

```
            $table->integer('before_quantity')->default(0);
```

```
            $table->integer('after_quantity')->default(0);
```

```
            $table->bigInteger('cost_price')->default(0);
```

```
            $table->text('notes')->nullable();
```

```
            $table->timestamps();
```

```
            $table->foreign('stock_movement_id')->references('id')->on('stock_movements')->cascade;
```

```
            $table->foreign('product_id')->references('id')->on('products');
```

```
            $table->foreign('product_variant_id')->references('id')->on('product_variants');
```

```
        });
```

```
    }
```

```
    public function down(): void { Schema::dropIfExists('stock_movement_items'); }
```

```
};
```

2. Enums

app/Modules/Inventory/Enums/MovementType.php

```
<?php

namespace App\Modules\Inventory\Enums;

enum MovementType: string
{
    case In = 'in'; // Barang masuk (manual/pembelian)
    case Out = 'out'; // Barang keluar (manual/kerusakan)
    case Transfer = 'transfer'; // Pindah gudang
    case Adjustment = 'adjustment'; // Penyesuaian stok (bisa in/out)
    case Opname = 'opname'; // Hasil stock opname (hitung fisik)
    case Sale = 'sale'; // Terjual via POS
    case Return = 'return'; // Retur pelanggan

    public function label(): string
    {
        return match ($this) {
            self::In => 'Barang Masuk',
            self::Out => 'Barang Keluar',
            self::Transfer => 'Transfer Antar Gudang',
            self::Adjustment => 'Penyesuaian Stok',
            self::Opname => 'Stock Opname',
            self::Sale => 'Penjualan',
            self::Return => 'Retur Pelanggan',
        };
    }

    public function isAddition(): bool
    {
        return in_array($this, [self::In, self::Return]);
    }

    public function isDeduction(): bool
    {
        return in_array($this, [self::Out, self::Sale]);
    }
}
```

app/Modules/Inventory/Enums/MovementStatus.php

```
<?php
```

```
namespace App\Modules\Inventory\Enums;
```

```
enum MovementStatus: string
```

```
{
```

```
    case Draft      = 'draft';
```

```
    case Pending    = 'pending';
```

```
    case Completed  = 'completed';
```

```
    case Cancelled  = 'cancelled';
```

```
    public function label(): string
```

```
    {
```

```
        return match ($this) {
```

```
            self::Draft      => 'Draft',
```

```
            self::Pending    => 'Menunggu',
```

```
            self::Completed  => 'Selesai',
```

```
            self::Cancelled  => 'Dibatalkan',
```

```
        };
```

```
    }
```

```
}
```

3. Models

app/Modules/Inventory/Models/Warehouse.php

```
<?php

namespace App\Modules\Inventory\Models;

use App\Modules\Tenant\Models\Branch;
use App\Shared\Traits\BelongsToBusiness;
use App\Shared\Traits\HasUuid;
use Illuminate\Database\Eloquent\Model;
use Illuminate\Database\Eloquent\Relations\BelongsTo;
use Illuminate\Database\Eloquent\Relations\HasMany;
use Illuminate\Database\Eloquent\SoftDeletes;

class Warehouse extends Model
{
    use HasUuid, BelongsToBusiness, SoftDeletes;

    protected $fillable = [
        'id', 'business_id', 'branch_id',
        'name', 'code', 'address',
        'is_main', 'is_active',
    ];

    protected function casts(): array
    {
        return [
            'is_main' => 'boolean',
            'is_active' => 'boolean',
        ];
    }

    /* — Relations — */

    public function branch(): BelongsTo
    {
        return $this->belongsTo(Branch::class, 'branch_id');
    }

    public function movements(): HasMany
```



```
{  
    return $this->hasMany(StockMovement::class, 'warehouse_id');  
}  
  
/* — Scopes — */  
  
public function scopeActive($query)  
{  
    return $query->where('is_active', true);  
}  
}
```

app/Modules/Inventory/Models/StockMovement.php

```
<?php
```

```
namespace App\Modules\Inventory\Models;
```

```
use App\Models\User;
```

```
use App\Modules\Inventory\Enums\MovementStatus;
```

```
use App\Modules\Inventory\Enums\MovementType;
```

```
use App\Shared\Traits\BelongsToBusiness;
```

```
use App\Shared\Traits\HasUuid;
```

```
use Illuminate\Database\Eloquent\Model;
```

```
use Illuminate\Database\Eloquent\Relations\BelongsTo;
```

```
use Illuminate\Database\Eloquent\Relations\HasMany;
```

```
use App\Shared\Helpers\CodeGenerator;
```

```
class StockMovement extends Model
```

```
{
    use HasUuid, BelongsToBusiness;

    protected $fillable = [
        'id', 'business_id', 'warehouse_id', 'user_id',
        'reference_number', 'type', 'status', 'notes',
        'reference_id', 'reference_type',
    ];

    protected function casts(): array
    {
        return [
            'type' => MovementType::class,
            'status' => MovementStatus::class,
        ];
    }

    protected static function boot(): void
    {
        parent::boot();

        static::creating(function (self $model) {
            if (empty($model->reference_number)) {
                $prefix = strtoupper(substr($model->type->value, 0, 3));
                $model->reference_number = CodeGenerator::invoiceNumber($prefix);
            }
        });
    }
}
```

```
        });  
    }  
  
    /* — Relations — */  
  
    public function warehouse(): BelongsTo  
    {  
        return $this->belongsTo(Warehouse::class, 'warehouse_id');  
    }  
  
    public function user(): BelongsTo  
    {  
        return $this->belongsTo(User::class, 'user_id');  
    }  
  
    public function items(): HasMany  
    {  
        return $this->hasMany(StockMovementItem::class, 'stock_movement_id');  
    }  
}
```

app/Modules/Inventory/Models/StockMovementItem.php

```
<?php

namespace App\Modules\Inventory\Models;

use App\Modules\POS\Models\Product;
use App\Modules\POS\Models\ProductVariant;
use App\Shared\Traits\HasUuid;
use Illuminate\Database\Eloquent\Model;
use Illuminate\Database\Eloquent\Relations\BelongsTo;

class StockMovementItem extends Model
{
    use HasUuid;

    protected $fillable = [
        'id', 'stock_movement_id', 'product_id', 'product_variant_id',
        'quantity', 'before_quantity', 'after_quantity',
        'cost_price', 'notes',
    ];

    protected function casts(): array
    {
        return [
            'quantity'          => 'integer',
            'before_quantity' => 'integer',
            'after_quantity'   => 'integer',
            'cost_price'       => 'integer',
        ];
    }

    /* — Relations — */

    public function stockMovement(): BelongsTo
    {
        return $this->belongsTo(StockMovement::class, 'stock_movement_id');
    }

    public function product(): BelongsTo
    {
        return $this->belongsTo(Product::class, 'product_id');
    }
}
```

```
public function variant(): BelongsTo
{
    return $this->belongsTo(ProductVariant::class, 'product_variant_id');
}
}
```